

Maharashtra HSC Exam, 2019

Physics

Time-3 Hours

M.M- 70

General Instructions:

1. All questions are compulsory.
2. Answers to the questions of Section-A, B, C and D should be written in the same answer book.
3. In Section –A there are 8 questions each of 1 marks.
4. In Section – B there are 7 questions each of 2 marks.
5. In Section- C there are 11 questions each of 3 marks.
6. In Section- D there are 3 questions each of 5 marks.
7. Draw neat, labelled diagrams and write balanced chemical equations wherever necessary.
8. Figures to the right indicate full marks.
9. Use of logarithmic table is allowed.
10. Every new question must be started on a new page.

SECTION A

1. When sparingly Solutionuble substance like alcohol is disSolutionved in water, surface tension of water
(A) Increases
(B) Decreases
(C) Remains constant
(D) Becomes infinite

Solution: B

Surface tension is decreases when sparingly Solutionuble substance are added.

2. The specific capacity of water is
(A) $8R$
(B) $\frac{7}{8}R$
(C) $9R$
(D) $\frac{9}{7}R$

Solution: C

One molecule of water is having 43 atoms.

Water can be considered as Solid as its volume is constant.

Specific heat capacity of one atom 8

Sold is $3R$

$3 \times 3R = 9R$

3. The electric field intensity outside the charged conducting sphere of radius 'R' placed in medium of permittivity ϵ at a distance 'r' from the center of the sphere in terms of surface charge density σ

(A) $\frac{\sigma}{\epsilon} \left(\frac{R}{r} \right)^2$

(B) $\frac{\sigma}{\epsilon} \left(\frac{r}{R} \right)^2$

(C) $\frac{\sigma}{\epsilon} \left(\frac{R^2}{r^2} \right)^2$

(D) $\frac{\epsilon}{\sigma} \left(\frac{r}{R} \right)^2$

Solution: A

By using Gauss theorem,

$$\oint \vec{E} d\vec{s} = q$$

$$\epsilon \oint ds = q$$

$$\epsilon E 4\pi r^2 = q$$

$$E = \frac{1}{4\pi \epsilon} \frac{q}{r^2}$$

$$\sigma = \frac{q}{4\pi R^2}$$

$$q = \sigma 4\pi R^2$$

$$E = \frac{1}{4\pi \epsilon} \frac{\sigma 4\pi R^2}{r^2}$$

$$= \frac{\sigma}{\epsilon} \left(\frac{R}{r} \right)^2$$

4. An electron of energy 150eV has wavelength of 10^{-10} m. The wavelength of a 0.60keV electron is,

(A) $0.50\text{\AA}$

(B) $0.750\text{\AA}$

(C) $1.2\text{\AA}$

(D) $1.5\text{\AA}$

Solution: A

$$\lambda = \frac{h}{\sqrt{meV}}$$

$$\lambda_1 \alpha = \frac{1}{\sqrt{v_1}}$$

$$\lambda_2 \alpha = \frac{1}{\sqrt{v_2}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{v_2}{v_1}}$$

$$\frac{10^{-10}}{\lambda_2} = \sqrt{\frac{0.60 \times 10^3}{1.50}}$$

$$\sqrt{\frac{600}{150}}$$

$$= 2$$

$$\lambda_2 = \frac{10^{-10}}{2}$$

$$0.5 \times 10^{-10}$$

$$0.5 \text{ \AA}$$

5. What is the value of tangential acceleration in U.C.M?

Solution:

The value of tangential acceleration in U.C.M is zero, due to its angular velocity is constant

$$\alpha = \frac{d\omega}{dt}$$

$$\omega = \text{const} \tan t$$

$$\alpha = 0$$

6. What happens to a ferromagnetic substance heated above Curie temperature?

Solution:

When ferromagnetic substance heated above Curie temperature it loses its magnetism and becomes paramagnetic substance.

$$X \propto \frac{1}{T}$$

7. At which position of the plane of the rotating coil with the direction of magnetic field, the e.m.f. induced in the coil is maximum?

Solution:

$$e = e_0 \sin \omega t$$

Where e_0 is peak value for e to be maximum.

$$\sin \omega t = 1$$

$$\omega t = \sin^{-1}[1] = 90^\circ$$

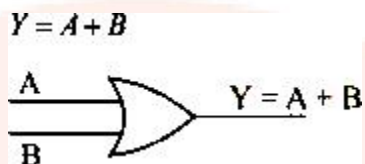
$$\omega t = 90^\circ$$

8. Name the logic gate which generated high output when at least input is high

Solution:

OR Gate

$$Y = A + B$$



A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

SECTION B

9. In Young's experiment interference bands were produced on a screen placed at 150cm from two slits, 0.15mm apart and illuminated by the light of wavelength 6500Å. Calculate the fringe width.

Solution:

We have,

$$D = 150\text{CM} = 150 \times 10^{-2} \text{m}$$

$$d = 0.15\text{MM} = 0.15 \times 10^{-3} \text{m}$$

$$\Lambda = 6500\text{\AA} = 6500 \times 10^{-10} \text{m}$$

$$X = ?$$

$$\therefore X = \frac{\lambda D}{d}$$

$$= \frac{6500 \times 15}{15} \times 10^{-12} 10^5$$

$$= 65 \times 10^{-7}$$

$$X = 6.5 \times 10^{-6} \text{m} = 6.5 \mu\text{m}$$

10. The susceptibility of magnesium at 300K is 1.2×10^{-5} . What will be its susceptibility at 200K.

Solution:

We have,

$$T_1 = 300K$$

$$X_1 = 1.2 \times 10^{-5}$$

$$T_2 = 200K$$

$$X_2 = ?$$

$$\therefore x \propto \frac{1}{T}$$

$$\therefore \frac{X_1}{X_2} = \frac{T_2}{T_1}$$

$$X_2 = \frac{X_1 \times T_1}{T_2}$$

$$= \frac{12 \times 10^{-5} \times 300}{200}$$

$$= 18 \times 10^{-6}$$

11. The length of second's pendulum in a clock is increased to 4 times its initial length. Calculate the number of oscillation completed by the pendulum in one minute.

Solution:

L_1 = initial length

T_1 = initial time period = 2s

L_2 = final length

T_2 = Final time period

$\therefore t_2 = 1 \text{ min} = 60\text{s}$

And

$$l_2 = 4l_1$$

$$\therefore T_1 = 2\pi \sqrt{\frac{l_1}{g}}$$

$$T_2 = 2\pi \sqrt{\frac{l_2}{g}}$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{l_1}{l_2}}$$

$$\Rightarrow \frac{2}{T_2} = \sqrt{\frac{l_1}{4l_1}}$$

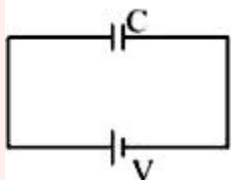
$$\Rightarrow \frac{2}{T_2} = \sqrt{\frac{1}{4}}$$

$$T_2 = \frac{1}{2}$$

OR

A body of mass 1kg is made to oscillate on a spring of force constant 16N/m. Calculate
(a) Angular frequency (b) Frequency of vibrations.

Solution:



We have,

$$m = 1\text{kg}$$

$$k = 16\text{N / m}$$

$$\omega = ?$$

$$n = ?$$

$$\therefore k = m\omega^2$$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

$$= \sqrt{\frac{16}{1}}$$

$$\therefore \omega = 4\text{rad / s}$$

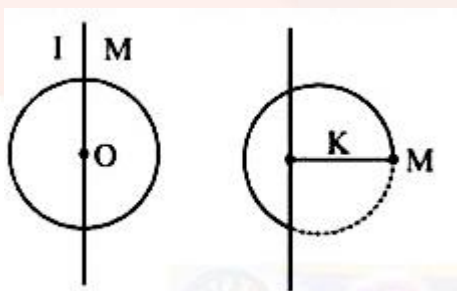
$$\beta\omega = 2\pi n$$

$$\therefore n = \frac{\omega}{2\pi} = \frac{4}{2 \times 3.14} = \frac{2}{3.14}$$

$$= 0.636\text{Hz}$$

12. Define capacitance of a capacitor and its SI unit.

Solution:



Capacitor is a device which is used to store charge and its capacity to store charge is known as its capacitance. Its SI unit is Farad.

13. Define radius of gyration. Write its physical significance.

Solution:

Radius of gyration is known as the perpendicular distance from the axis of rotation at which entire mass of the body supposed to concentrate so as to have the same moment of inertia as that of the body.

Figure

$$I = MK^2$$

$$K^2 = \frac{I}{M}$$

$$K = \sqrt{\frac{I}{M}}$$

Its physical signification is, it signifies distribution of mass along axis of rotation.

14. Distinguish between p-type and n-type semiconductor

Solution:

n-type	p-type
(i) Number of electrons is more than number of holes.	(a) Number of holes are more than number of electrons
(ii) Pentavalent impurities are used for doping	(b) Trivalent impurities are used for doping
(iii) These are called donors	(c) These are called acceptors
(iv) Electrons are minority charge carriers.	(d) Holes are majority charge carriers.

15. Explain the terms (a) Transducer and (b) Attenuation in communication system.

Solution:

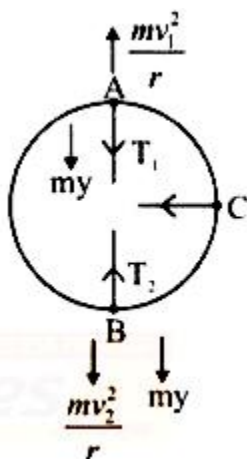
(a) Transducer: - Transducer is the device which converts one form of energy into another.

(b) Attenuation: - Loss strength of signals while propagation through medium is known as attenuation.

SECTION C

16. Obtain expression of energy of a particle at different positions in the vertical motion.

Solution:



At A

$$T_1 + mg = \frac{mv_1^2}{r}$$

$$\text{Let, } T_1 = 0$$

$$mg = \frac{mv^2}{r}$$

$$v_1 = \sqrt{rg} \dots (1)$$

At B

$$\Delta PE_A + \Delta KE_A + \Delta KE_B$$

$$(mg2r) + \frac{1}{2}mv_1^2 = mg(0) + \frac{1}{2}mv_2^2$$

$$2mgr + \frac{1}{2}m(\sqrt{rg})^2 = 0 + \frac{1}{2}mv_2^2$$

$$4rg + rg = 2gr + v_3^2$$

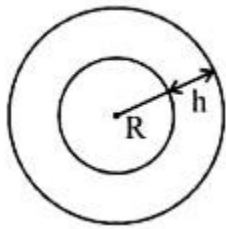
$$v_3^2 = 3rg$$

$$v_3 = \sqrt{3rg}$$

17. Define binding energy and obtain an expression for binding energy of a satellite revolving in a circular orbit round the Earth.

Solution:

We know that the product of gravitational potential and mass is equal to binding energy



$$-\frac{GMm}{(r+h)} \times m$$

$$= -\frac{GMm}{(r+h)}$$

$$\therefore E = \frac{1}{2} mV^2$$

$$\frac{1}{2} m \left(\sqrt{\frac{GM}{(r+h)}} \right)^2$$

$$= \frac{mGM}{2(r+h)}$$

$$T.E = P.E + K.E$$

$$= -\frac{GMm}{(r+h)} + \frac{GMm}{2(r+h)}$$

$$= \frac{GMm}{(r+h)} \left(-1 + \frac{1}{2} \right)$$

$$= -\frac{GMm}{2(r+h)}$$

18. State Hooke's law. Define elastic limit and modulus of elasticity.

Solution:

Hooke's law:- It is defined as the force required to compress or elongate the spring by a small

Distance is proportional to that distance.

$$\sigma = -K\varepsilon$$

Where, σ is the applied force

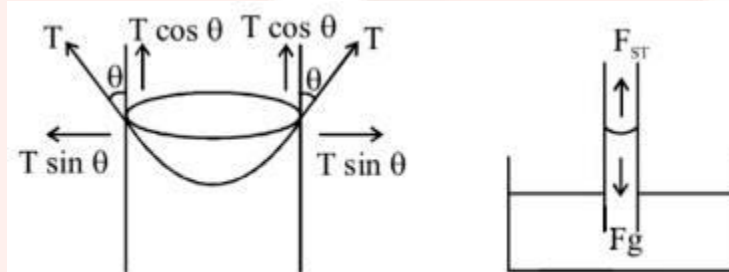
ε Is the strain

K is the constant

Elastic limit: The maximum value of stress where stress is directly proportional to strain is known as elastic limit.

19. Obtain an expression for the rise of a liquid in a capillary tube.

Solution:



$$T = \frac{\text{Force due to surface tension}}{\text{length}}$$

$$\therefore T \cos \theta = \frac{F_{ST}}{2\pi r}$$

$$\Rightarrow F_{ST} = T \cos \theta \times 2\pi r$$

$\therefore$ at equilibrium,

$$T \cos \theta \times 2\pi r = mg$$

$$\left(\rho = \frac{\text{mass}}{\text{volume}} = \frac{m}{\pi r^2 h} \right)$$

$$(m = \rho \pi r^2 h)$$

$$T \cos \theta \times 2\pi r = \rho \pi r^2 h g$$

$$\Rightarrow h = \frac{2T \cos \theta}{\rho g}$$

20. Explain the reflection of transversal and longitudinal waves from a denser medium and rarer medium.

Solution:

Reflection of transverse wave:

We know that when the transverse wave is travelling from denser to rarer media crest is reflected as crest and trough is reflected as trough such that phase change is zero.

While,

Reflection of longitudinal wave:

When the transverse wave is reflected as rarefaction and rarefaction is reflected as compression, such that phase difference of π .

21. What is the photo electric effect? Define (i) stopping potential (ii) Photoelectric work function.

Solution:

The process of emission of electron by a certain substance, when it is exposed to radiation of suitable frequencies is known as photo electric effect. And the emitted electrons are known as photoelectrons.

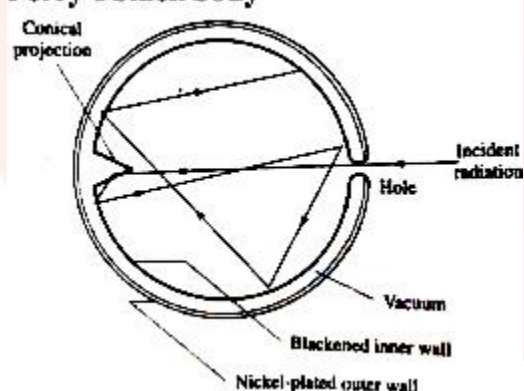
- (i) It is the minimum negative potential given to the plate for which photoelectric current stops or becomes zero.
- (ii) It is the minimum energy required to remove an electron from a given surface is known as the photo electric work function.

22. What is perfectly black body? Explain Ferry's black body.

Solution:

A body which absorbs the entire energy incident on it is known as perfectly black body. And the absorption of lamp black is 98%.

Ferry's black body



- (i) It consists of double walled hollow metal sphere having small aperture.
- (ii) Inner surface is coated with lamp black.
- (iii) Outer surface is Nickel polished
- (iv) The space between spheres is evacuated.

- (v) And the energy entering suffers multiple reflection and get 98% absorbed at each reflection.
- (vi) Hence, the aperture acts as perfectly black body.

23. When a resistor of 5Ω is connected across the cell, its terminal potential difference is balanced by 150cm of potentiometer wire and when a resistance of 10Ω is connected across the cell, the terminal potential difference is balanced by 175cm same potentiometer wire. Find the balancing length when the cell is in open circuit and the internal resistance of the cell.

Solution:

$$r = 5 \left(\frac{l_1}{150} - 1 \right) \dots\dots\dots (1)$$

$$r = 10 \left(\frac{l_1}{175} - 1 \right) \dots\dots\dots (2)$$

$$5 \left(\frac{l_1}{150} - 1 \right) = 10 \left(\frac{l_1}{175} - 1 \right)$$

$$\frac{l_1}{150} - 1 = \frac{2l_1}{175} - 2$$

$$-1 + 2 = \frac{2l_1}{175} - \frac{l_1}{150}$$

$$1 = l_1 \left(\frac{2}{175} - \frac{1}{150} \right)$$

$$= l_1 \left(\frac{300 - 125}{150 \times 175} \right)$$

$$\Rightarrow l_1 = 210 \text{ cm}$$

$$r = R \left(\frac{l_1}{l_2} - 1 \right)$$

$$= 5 \left(\frac{210}{150} - 1 \right)$$

$$= 2\Omega$$

24. A cyclotron is used to accelerate protons to a kinetic energy of 5MeV. If the strength of the magnetic field in the cyclotron is 2T, find the radius and the frequency needed for applied alternating voltage of the cyclotron.

Solution:

$$K.E = 5MeV$$

$$B = 2T$$

$$r = ?$$

$$f = ?$$

$$v = 3 \times 10^7 \text{ m/s}$$

$$r = \frac{mv}{Bq} = \frac{1.67 \times 10^{-27} \times 3 \times 10^7}{2 \times 1.6 \times 10^{-19}}$$

$$= 0.1565m$$

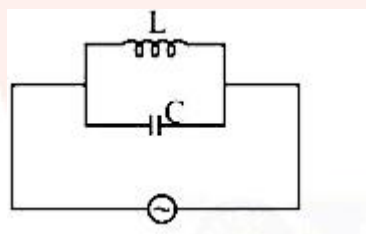
$$f = \frac{qB}{2\pi m}$$

$$= \frac{1.6 \times 10^{-19} \times 2}{2 \times 3.14 \times 1.67 \times 10^{-27}}$$

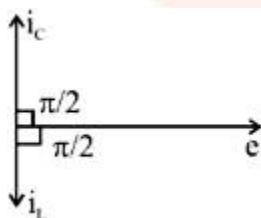
$$= 3.05 \times 10^7 \text{ Hz}$$

25. Assuming expression for impedance in a parallel resonant circuit, state the condition for parallel resonance. Define resonant frequency and obtain an expression for it.

Solution:



$$i = i_0 \sin \omega t$$



$$i = i_c - i_L$$

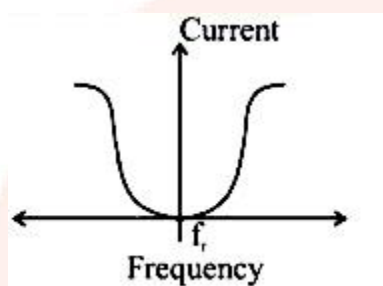
$$\frac{e}{z} = \frac{e}{X_c} - \frac{e}{X_L}$$

And for parallel resonance

$$X_L = X_C$$

$$\frac{1}{z} = 0$$

$$z = \infty$$



Resonant frequency (f_r),

It is the frequency at which impedance (Z) becomes infinity and current becomes zero is known as resonant frequency.

$$\omega L = \frac{1}{\omega C}$$

$$\omega^2 LC = 1$$

$$\omega^2 = \frac{1}{LC}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$2\pi f_r = \frac{1}{\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

26. Using an expression for energy of electron, obtain the Bohr's formula for hydrogen spectral lines.

Solution:

Let us suppose E_n and E_p be the energies of electron in n^{th} and p^{th} orbit respectively.

$$\therefore \text{Energy of electron in } n^{\text{th}} \text{ or orbit is } E_n = -\left(\frac{me^4}{8\varepsilon_0^2 h^2}\right) \frac{1}{n^2} \dots (1)$$

$$\therefore \text{Energy of electron in } p^{\text{th}} \text{ or orbit is } E_p = -\left(\frac{me^4}{8\varepsilon_0^2 h^2}\right) \frac{1}{p^2} \dots (2)$$

$\therefore$ A/c to Bohr's postulate energy emitted is,

$$\therefore \text{energy emitted, } h\nu = E_n - E_p$$

$$\therefore h\nu = \frac{me^4}{8\varepsilon_0^2 h^2} \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$$

$$\Rightarrow \nu = \frac{me^4}{8\varepsilon_0^2 h^3} \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$$

$$\therefore \frac{1}{\lambda} = R \left(\frac{1}{p^2} - \frac{1}{n^2} \right) \left[\because \nu = \frac{c}{\lambda} \right]$$

And,

$$R = \frac{me^4}{8\varepsilon_0^2 h^2}$$

OR

State the law of radioactive decay. Hence derive the relation $N = N_0 e^{-\lambda t}$, represent it graphically.

Solution:

Let us suppose N be the number of nuclei present at any instant 't'

And 'dN' be the number of nuclei that disintegrated in short interval of time 'dt'

Therefore,

$$\frac{-dN}{dt} \propto N$$

$$\therefore \frac{dN}{dt} = -\lambda N \quad \dots(1)$$

Where, λ is decay constant

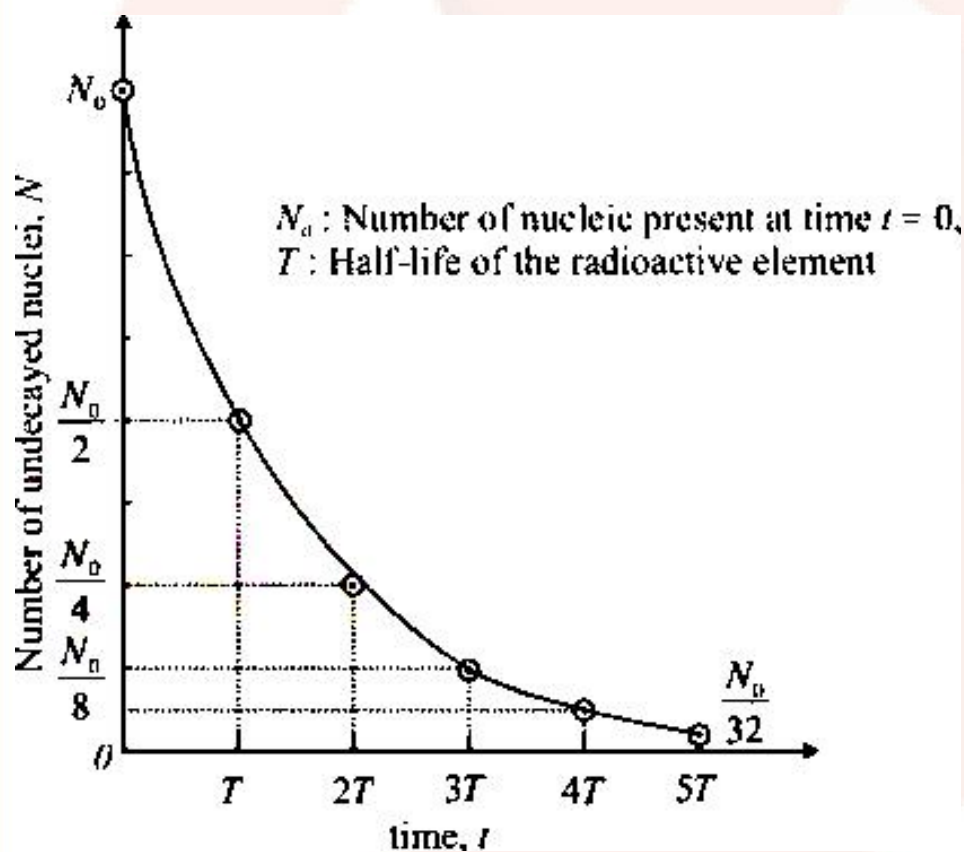
$$\therefore \frac{dN}{N} = -\lambda dt$$

by integrating both sides,

$$\int \frac{dN}{N} = \int -\lambda dt$$

$$\ln N = -\lambda t + C \quad (2)$$

Where, C is the integration constant.



$$\text{at } t=0, N=N_0$$

$$\therefore \ln N_0 = C \dots (3)$$

from eq(2) and (3)

$$\ln N = -\lambda t + \ln N_0$$

$$\lambda t = \ln N - \ln N_0$$

$$-\lambda t = \ln \left(\frac{N}{N_0} \right)$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$N = N_0 e^{-\lambda t}$$

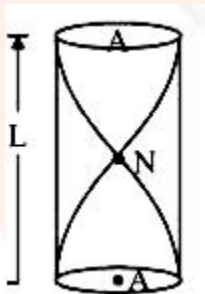
SECTION D

27. Show that even as well as odd harmonics are present as overtones in the case of an air column vibrating in a pipe open at both the ends.

A wheel of moment of inertia 1kg.m^2 is rotating at a speed of 30rad/s . Due to friction on the axis, it comes to rest in 10 minutes. Calculate the average torque of the friction.

Solution:

Mode 1



$$L = \frac{\lambda}{2}$$

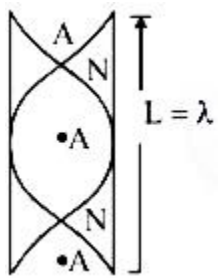
$$\lambda = 2L$$

$$v = n\lambda$$

$$n = \frac{v}{\lambda} = \frac{v}{2L}$$

$$\therefore \text{fundamental frequency } n = \left(\frac{v}{2L} \right)$$

Mode 2



$$v = n_1 \lambda$$

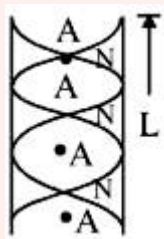
$$n_1 = \frac{V}{\lambda_1}$$

$$\lambda_1 = L$$

$$n_1 = \frac{V}{L}$$

$$h_1 = 2 \left(\frac{V}{2L} \right)$$

Mode 3



$$L = \frac{2\lambda_1}{3}$$

$$V = n_2 \lambda_2$$

$$n_2 = \frac{V}{\lambda_2}$$

$$n_2 = 3 \left[\frac{V}{2L} \right]$$

$$n_2 = 3n$$

Now, in general

$$n_p = (P+1)n$$

$$(P+1)\frac{V}{2L}$$

$$P = 1, 2, 3, \dots$$

Now, we have

$$I = 1 \text{ kg m}^2$$

$$\omega_0 = 30 \text{ rad / s}$$

$$\omega = 0$$

$$t = 10 \text{ min} = 10 \times 60 = 600 \text{ s}$$

$$\alpha = \frac{\omega - \omega_0}{600}$$

$$= \frac{0 - 30}{600}$$

$$-0.05 \text{ rad / s}^2$$

$$\tau = I\alpha$$

$$= 0.05 \times I$$

$$= -0.05 \text{ Nm}$$

$$\theta = \omega t + \frac{1}{2} \alpha t^2$$

$$= 30 \times 600 - \frac{1}{2} 0.05 \times 300 \times 60 \text{ rad}$$

$$= 18000 - 180000 \times 0.5$$

$$18000 - 9000$$

$$\theta = 9000$$

$$\therefore \omega = \tau \theta$$

$$= 0.05 \times 9000$$

$$= -450 \text{ J}$$

OR

Explain the formation of stationary waves by analytical method. Show that nodes and antinodes are equally spaced in stationary waves.

The radius of gyration of a body about an axis, at a distance of 0.4m from its center of mass is 0.5m. Find its radius of gyration about a parallel axis passing through its center of mass.

Solution:

Analytical treatment of Stationary Waves:,

Let us consider two simple harmonic progressive waves having same amplitude(a) frequency(n) and period (T) travelling along same medium.

$$Y_1 = a \sin 2\pi \left(nt - \frac{x}{\lambda} \right) \dots (1)$$

And wave along -ve direction

$$Y_2 = a \sin 2\pi \left(nt + \frac{x}{\lambda} \right) \dots (2)$$

$$\therefore Y = Y_1 + Y_2$$

$$Y = a \sin 2\pi \left(nt - \frac{x}{\lambda} \right) + a \sin 2\pi \left(nt + \frac{x}{\lambda} \right)$$

$$2a \sin \frac{2\pi}{2} \left(nt - \frac{x}{\lambda} + nt + \frac{x}{\lambda} \right) \cdot \cos \frac{2\pi}{2} \left(nt - \frac{x}{\lambda} - nt - \frac{x}{\lambda} \right)$$

$$\therefore Y = 2a \sin 2\pi nt \cdot \cos \left(\frac{-x}{\lambda} \right)$$

$$= 2a \left[\sin (2\pi nt) \cdot \cos \left(\frac{2\pi x}{\lambda} \right) \right]$$

$$= \left[2a \cos \left(\frac{2\pi x}{\lambda} \right) \right] \sin (2\pi nt)$$

$$= A \sin (2\pi nt)$$

$$\text{Where, } A = 2a \cos \left(\frac{2\pi x}{\lambda} \right)$$

Now, the condition for antinode,

The points of medium which vibrate with maximum amplitude

$$A = \pm 2a$$

$$\therefore 2a \cos \frac{2\pi x}{\lambda} = \pm 2a$$

$$\Rightarrow \cos \frac{2\pi x}{\lambda} = \pm 1$$

$$\frac{2\pi x}{\lambda} = 0, \pi, 2\pi, \dots$$

$$\Rightarrow \frac{2\pi x}{\lambda} = P\pi$$

$$\therefore x = \frac{\lambda P}{2}$$

$$\frac{2\pi x}{\lambda} = \pi, x = \frac{\lambda}{2}$$

$$\frac{2\pi x}{\lambda} = 2\pi, x = \lambda$$

$$\therefore x = 0, \frac{\lambda}{2}, \lambda, \dots$$

Condition for nodes,

The points of medium which vibrate with minimum amplitudes are called nodes.

$$A = 0$$

$$\Rightarrow 2a \cos \left(\frac{2\pi x}{\lambda} \right) = 0$$

$$\cos \frac{2\pi x}{\lambda} = 0$$

$$\frac{2\pi x}{\lambda} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\text{If } \frac{2\pi x}{\lambda} = \frac{\pi}{2}, x = \frac{\lambda}{4}$$

$$\frac{2\pi x}{\lambda} = \frac{3\pi}{2}, x = \frac{3\lambda}{4}$$

$$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$$

Now, we have

$$h = 0.4m$$

$$K = 0.5m$$

$$K_{cm} = ?$$

By the theorem of parallel axis,

$$I = I_C + Mh^2$$

$$I = MK^2$$

$$\Rightarrow K^2 = K_{CM}^2 + h^2$$

$$\Rightarrow (0.5)^2 = (0.4)^2 + K_{CM}^2$$

$$\Rightarrow K_{CM}^2 = (0.5)^2 - (0.4)^2$$

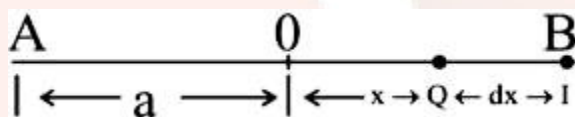
$$= 0.09$$

$$K_{CM} = 0.3m$$

28. Obtain an expression for potential energy of a particle performing S.H.M. What is the value of potential energy (i) Mean position, and (ii) Extreme position.

A stretched sonometer wire is in unison with a tuning fork. When the length of the wire is increased by 5% the number of beats heard per second is 10. Find the frequency of the tuning fork.

Solution:



Let us consider a particle of mass 'm' performing S.H.M along AB about mean position 'O'

$$OB = OA = a$$

Let $F = kx$ be the restoring force, k is the force constant.

Therefore, work done

$$dw = -fdx = -(-kx)dx$$

$$= kx dx$$

$\therefore$ total workdone,

$$\int dw = \int_{x=0}^x kx dx = k \int_{x=0}^x x dx$$

$$\Rightarrow W = \frac{1}{2} kx^2 = \frac{1}{2} m\omega^2 x^2 \quad \left[\because \omega^2 = \frac{k}{m} \right]$$

$$\therefore P.E = \frac{1}{2} kx^2 = \frac{1}{2} m\omega^2 x^2$$

At mean position: $-x = 0$
 $\therefore P.E = 0$

At extreme position

$$x = \pm a$$

$$\therefore P.E = \frac{1}{2} m \omega^2 a^2$$

Since, we have

$$\frac{L_1}{L_2} = \frac{100}{105}$$

$$\therefore L_2 = 1.05 L_1$$

$$\because L_2 > L_1, n_2 < n_1$$

$$n_1 - n_2 = 10 \quad 7$$

Now, by the law of length vibrating string

OR

From differential equation of linear S.H.M, obtain an expression for acceleration, velocity and displacement of particle performing S.H.M.

A sonometer wire 1 m long weighing 2g is in resonance with a tuning fork of frequency 300Hz. Find tension in the sonometer wire.

Solution:

The differential equation of linear S.H.M is

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

where m=mass

$$\frac{d^2 x}{dt^2} = \text{acceleration of the particle}$$

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

Let m

$$\frac{k}{m} = \omega^2,$$

$$\therefore \frac{d^2x}{dt^2} + \omega^2x = 0$$

$$\text{Acceleration} = \frac{d^2x}{dt^2} = -\omega^2x \quad \dots\dots(1)$$

-ve sign shows acceleration and the displacement are in opposite direction.

$$a = \frac{d^2x}{dt^2} = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt}$$

$$= \frac{dv}{dx} \cdot v \quad \left[\because v = \frac{dx}{dt} \right]$$

$$\therefore v \frac{dv}{dx} = -\omega^2x$$

$$\Rightarrow vdv = -\omega^2x dx$$

Now, integrating both sides then we get,

$$\frac{v^2}{2} = \frac{-\omega^2x^2}{2} + C$$

C is the boundary condition.

When, $x = \pm A$, then

$$\therefore 0 = \frac{-\omega^2A^2}{2} + C$$

$$\Rightarrow C = \frac{\omega^2A^2}{2}$$

$$\therefore \frac{v^2}{2} = \frac{-\omega^2x^2}{2} + \frac{-\omega^2A^2}{2}$$

$$\Rightarrow v = \pm \omega \sqrt{A^2 - x^2} \quad \dots\dots(2)$$

This shows the velocity of the particle in the terms of displacement

Since,

$$\therefore v = \frac{dx}{dt}$$

$\therefore eq(2)$ becomes

$$\frac{dx}{dt} = \omega \sqrt{A^2 - x^2}$$

$$\Rightarrow \frac{dx}{\sqrt{A^2 - x^2}} = \omega dt$$

By integrating both sides we get,

$$\sin^{-1}\left(\frac{x}{A}\right) = \omega t + \alpha \dots\dots\dots(3)$$

Where, α is the constant of integration

$$\frac{x}{A} = \sin(\omega t + \alpha)$$

Therefore, displacement as a function of time is,

$$X = A \sin(\omega t + \alpha)$$

$$\therefore L = 1m, \text{mass} = 2g = 0.002kg, n = 300Hz$$

$$= \frac{0.002}{1} = 0.002kg / m$$

$$n = \frac{1}{2L} \sqrt{\frac{T}{m}}$$

$$\therefore T = 4n^2 L^2 m$$

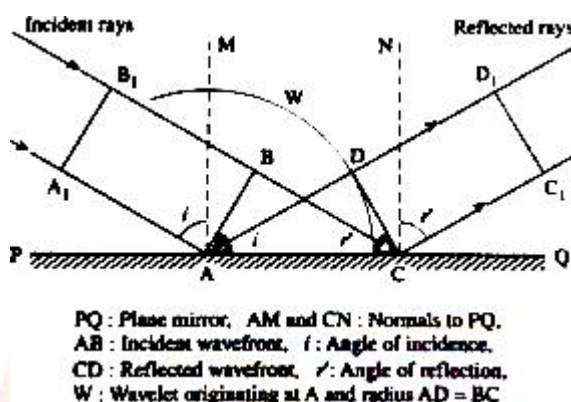
$$= 4 \times (300)^2 \times (1)^2 \times 0.002$$

$$= 720N$$

29. Explain refraction of light on the basis of wave theory. Hence prove the laws of refraction. Two coherent sources of light having intensity ratio 81:1 produce interference fringes. Calculate the ratio of intensities at the maxima and minima in the interference pattern.

Solution:

Let us consider a plane wave front of monochromatic light incident obliquely on a plane mirror PQ.



A_1B_1 and AB are the successive position of their incident wave front one wavelength apart,
 A_1A and B_1B are the corresponding incident rays of light.

If t is the time taken by the incident wave front to travel B to C, then $BC = vt$. At this time secondary waves originating at A cover the same distance, such that the necessary spherical wavelets has a radius vt .

If a normal AN is drawn to PQ,

$\angle A_1AM = i =$ angle of incidence and $\angle MAD = r =$ angle of reflection.

In $\triangle ABC$ and $\triangle ADC$,

$$AD = BD = vt$$

$$\angle ABC = \angle ADC = 90^\circ$$

$\therefore \triangle ABC$ and $\triangle ADC$ are congruent.

$$\Rightarrow \angle BAC = \angle DCA \quad \dots\dots(1)$$

But,

$$\angle A_1AM + \angle MAB = 90^\circ$$

$$\therefore \angle BAC = \angle DCA = i \quad \dots(2)$$

Also,

$$\angle MAD + \angle DAC = 90^\circ$$

$$\angle DAC + \angle DCA = 90^\circ$$

$$\therefore \angle DCA = \angle MAD = r \quad (3)$$

Therefore, eq (1) (2) and (3) we get,

$$i = r$$

So, the angle of incidence is equal to the angle of reflection. And the incident ray and reflected ray lie on the opposite sides of the normal of the reflecting surface at the point.

$$I_1 : I_2 = 81:1$$

Here, A_1 and A_2 are the amplitudes of interfering waves, such that the ratio of the intensity maximum to the intensity minimum in the fringe system is:

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{A_1 + A_2}{A_1 - A_2} \right)^2 = \left(\frac{r+1}{r-1} \right)^2$$

$$\text{Where, } r = \frac{A_1}{A_2}$$

$$\frac{I_1}{I_2} = \left(\frac{A_1}{A_2} \right)^2 = r^2$$

$$\therefore r = \sqrt{\frac{I_1}{I_2}} = \sqrt{81} = 9$$

$$\begin{aligned} \therefore \frac{I_{\max}}{I_{\min}} &= \left(\frac{9+1}{9-1} \right)^2 = \left(\frac{10}{8} \right)^2 \\ &= \frac{25}{16} \end{aligned}$$

Hence, the ratio of intensities of maxima and minima in the fringe system is 25:16.

OR

State Brewster's law and show that when light is incident at polarizing angle the reflected and refracted rays are mutually perpendicular to each other.

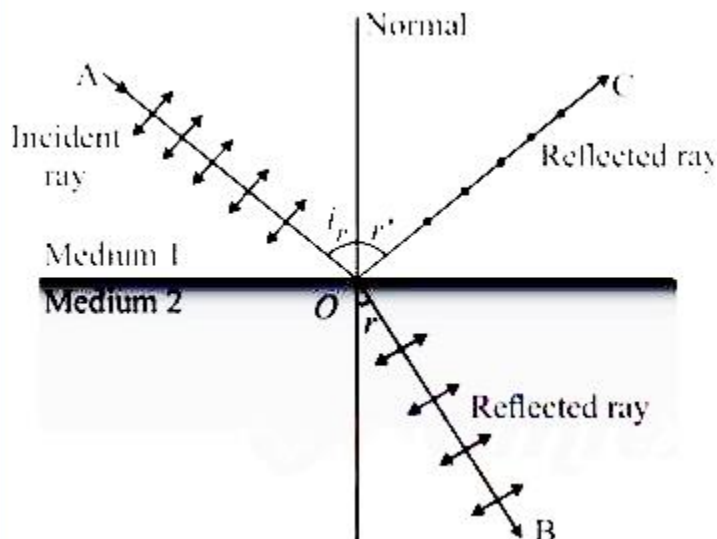
Monochromatic light of wavelength $4300\text{\AA}$ falls on a slit of width 'a'. for what value of 'a' the first maximum falls at 30° ?

Solution:

Brewster's law:- It is defined As the tangent of polarizing angle is equal to the refractive index of the reflecting medium with respect to the surrounding (n_2).

$$\tan i_p = n_2 = \frac{n_2}{n_1} \dots (1)$$

Where, n_1 is the absolute refractive index of the surrounding and n_2 is that of reflecting medium.



Here, a ray AO of unpolarized light incident on the interface separating two mediums. And the degree of polarization of the reflected ray OC varies with the angle of incidence and is a maximum for the angle of incidence is equal to i_p .

$$\frac{n_2}{n_1} = \frac{\sin i_p}{\sin r_p} \dots\dots\dots (2)$$

From eq (1) and (2)

$$\tan i_p = \frac{\sin i_p}{\cos r_p} = \frac{\sin i_p}{\sin i_r}$$

$$\therefore \cos i_p = \sin r$$

Now,

$$r' = i_p$$

$$\therefore \cos r' = \sin r$$

$$\Rightarrow \cos r' = \sin r$$

$$\Rightarrow \sin (90 - r') = \sin r$$

$$\Rightarrow r = 90^\circ - r'$$

$$\Rightarrow r' + r = 90^\circ$$

$$\therefore \angle COB = 90^\circ$$

This shows that complete polarization of the reflected ray at the polarizing angle, the reflected and refracted are mutually perpendicular

$$\lambda = 4300 \text{ \AA}$$

$$= 4300 \times 10^{-10} \text{ m}$$

$$a = ?$$

$$n = 1$$

$$\sin \theta = \frac{\lambda}{a}$$

$$\therefore a = \frac{\lambda}{\sin 30^\circ} = \frac{4300 \times 10^{-10}}{\frac{1}{2}}$$

$$a = 4300 \times 10^{-10} \times 2$$

$$= 8600 \times 10^{-10} \text{ m}$$

for first maximum

$$\sin \theta = \left(n + \frac{1}{2} \right) \frac{\lambda}{a}$$

$$= \frac{3}{2} \times \frac{4300 \times 10^{-10}}{\frac{1}{2}}$$

$$= 1.29 \mu\text{m}$$